Will Humphlett

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Experience

Roblox · San Mateo, CA
Data Platform

Senior Software Engineer Aug 2025 – Present

- Rebuilt event ingest from scratch as an in-house Kinesis and Firehose, with client libraries in five languages. At over 10 trillion events a day, its custom Go gateway performs hard schema validation and writes to Kafka for online consumers, and custom collectors move events from Kafka into the S3 data lake for offline analytics. Cut ingest costs by over two-thirds compared with the legacy system, which saves $10M a year.
- Built a typed runtime config service by rebuilding git on a key-value store. Its configs can be code, so teams change sampling logic on game servers live instead of waiting out the engine's three-week release cycle.
- Led the cross-org push to bring protobuf and gRPC into the C++ game engine on desktop, mobile and consoles, so any engine code can make typed calls to backend services. Moving stats telemetry from uncompressed query strings to protobuf cut its bandwidth by 90%.
- Led GDPR compliance for Roblox's event data with Legal and Infosec. Across about 6,000 tables, every schema must declare its PII, and the gateway pseudonymizes it before events reach Kafka.
- Designed the successor to Roblox's batch platform, which runs 60,000 Spark jobs a day across 17 clusters and 13 Airflow deployments. It is slated to replace them with one platform that gives each team guaranteed capacity and its own cost tracking. All 10,600 pipelines will share one asset graph, with cross-team dependencies, type checks, lineage and SLA alerts.

Software Engineer Aug 2023 – Aug 2025

- Wrote the first pipelines behind the engagement, retention, monetization and acquisition metrics every Roblox experience gets automatically. HyperLogLog and Theta sketches keep their distinct counts fast at any slice for 17.7 million queries a day.
- Built the pipeline behind Roblox's custom analytics, from the engine's Luau API to serving from Druid. At 32.6 billion economy, funnel and custom events a day, it serves 8 of the top 10 experiences, and its ingest path grew into Roblox's main event ingest.
- Designed and built Roblox's spatial analytics, starting with player heatmaps for every session. Aggregating on the client and again on the server cut a projected 187 TB a day to 30 GB.

Software Engineering Intern May 2022 – Dec 2022

ADTRAN · Huntsville, AL

Cooperative Education Student, three terms Jan 2020 – Aug 2021

Skills

Languages: Go (Golang), Python, C++, Java, TypeScript, Luau, SQL, Protobuf, gRPC
Data: Kafka, Flink, Spark, Airflow, Dagster, Druid, Trino, Parquet, S3, Redis, Data sketches
Infra: Kubernetes, AWS, Terraform, Prometheus, Grafana, GitHub Actions, CMake, Buck2, Bazel

Education

Bachelor of Software Engineering, Auburn University · GPA 3.86 May 2023